

ICOM IC-2730A Compact Organizer

RIDGID Pro Gear 2.0 build plan • mobile install, GMC Sierra 1500

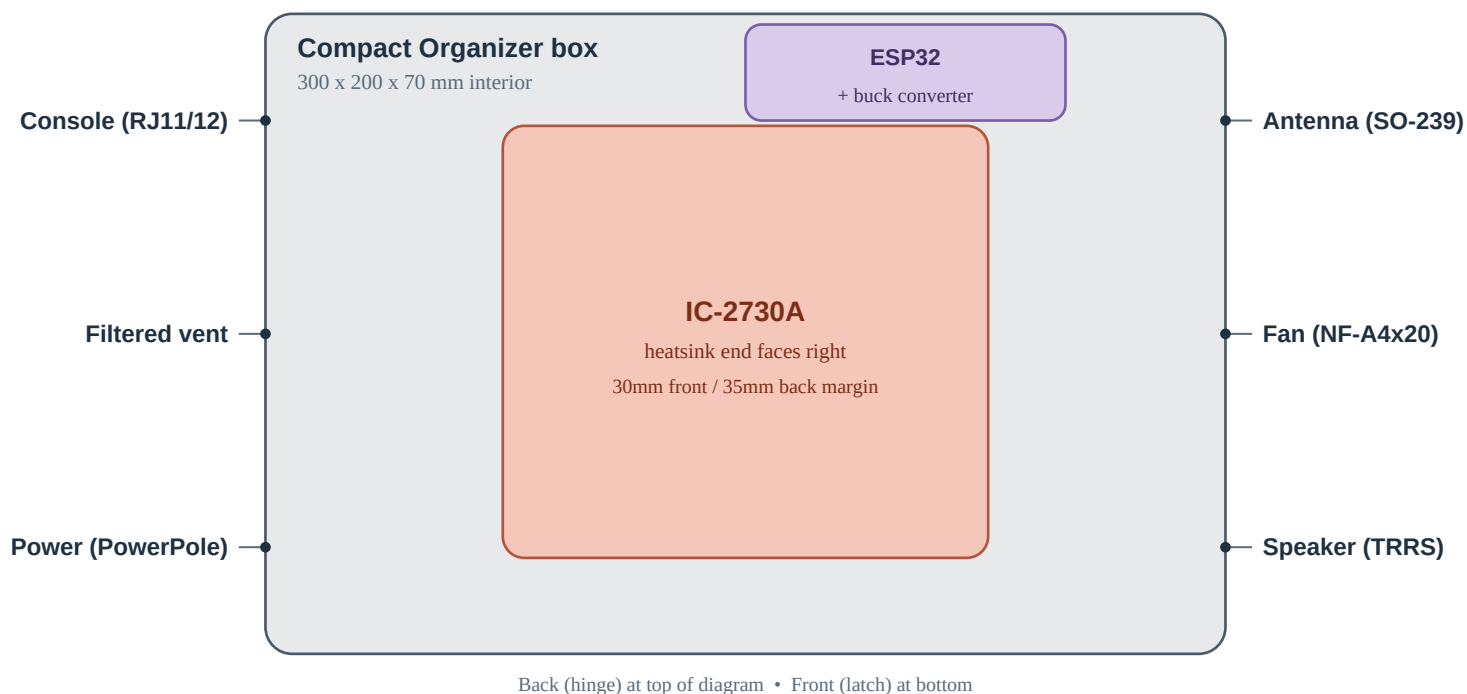
1. Design Overview

This box houses the ICOM IC-2730A main RF unit (control head mounted separately in the cab) inside a RIDGID Pro Gear 2.0 “Half Organizer” (254071), interior approximately 300 × 200 × 70 mm. The box sits stacked directly on top of the battery box, so only the left and right walls are usable for connector cutouts — the top is the lid and the back wall is against the stack.

Cooling is vented forced air: a filtered intake on the left wall and a Noctua fan exhausting on the right, pulling air directly across the radio. The IC-2730A has its own fan at the heatsink end, and the case fan sits directly behind both it and the adjacent heatsink — so the case fan's job is to carry that discharge out of the box, not to drive air through the fin channels itself. That is why the case fan's height is not tied to the heatsink's fin-channel direction. A sealed conduction-plate approach was ruled out because the box's floor sits flush on the battery box below it — there is no ambient air on the far side of the floor for heat to go.

Power and control cabling (DC power, console/control-head cable) route out the left wall; antenna and speaker audio route out the right wall. This keeps DC and switching noise away from the RF and audio paths, consistent with the noise-isolation practice used elsewhere in the build (ferrite choke at the DC entry, twisted/separated runs). There is no master power switch — the PowerPole disconnect already serves that role, and a switch on this box would sit out of reach of the operating position.

Floor Plan



2. Parts & Hole Schedule

Component	Wall / Position	Key Dimensions	Notes
SO-239 bulkhead	Right wall, back row	16 mm hole, 25×25 mm flange, 22 mm depth	Chassis-mount UHF female, solder-cup
TRRS bulkhead	Right wall, front row	22 mm dia., 30 mm depth	Speaker output
Fan — Noctua NF-A4x20 5V PWM	Right wall, center	40×40×20 mm, 32×32 mm hole spacing	4-pin true PWM, powered from 5V buck rail, not raw 12V
Filtered vent	Left wall, center	40×40mm frame · 32×32mm hole spacing (provisional — matches the fan footprint)	Media: 120 μm polyester monofilament mesh, washable, clamped between shroud and the outside face of the wall so it cleans without opening the box. TPU-printed shroud, ~95A shore hardness; one shroud design can serve both fan and vent; actual open-air area still depends on shroud design
PowerPole bulkhead	Left wall, front row	34.2×17.2mm flange · 19.1×11.2mm through-box cutout · 2× Ø 3.7mm holes, 26.7mm apart	Printed tower mount, flange-flush — connector body hides inside the box, not proud of the wall. Doubles as the box's power disconnect
Console bulkhead (RJ11/12)	Left wall, back row	21 mm round · flat side to 19 mm — drill round, file the flat	Weatherproof, tethered threaded cap, pigtail lead (not a straight pass-through)
ESP32 dev board	Floor, back wall, ~50 mm in from right, 100 mm wide	—	Fan PWM control, tach (with divider), DS18B20 read, temp logging
Buck converter (12V→5V)	Same shelf as ESP32	Automotive-transient rated	Feeds ESP32 and fan; not the radio
DS18B20 temp probe	Mounted against heatsink	Waterproof probe form	1-Wire, 4.7kΩ pull-up to 3.3V

Wiring reference

Run	Gauge	Notes
12V main (radio)	12 AWG	Anderson PowerPole; ferrite choke at DC entry; ~13A TX draw

Run	Gauge	Notes
5V rail (ESP32 + fan)	22 AWG	Off buck converter output, not raw 12V
Fan PWM control	—	ESP32 GPIO direct, 3.3V logic, no MOSFET needed
Fan tach (optional)	—	10k/20k divider before ESP32 GPIO — fan's internal pull-up is 5V
DS18B20 data	—	4.7k Ω pull-up to 3.3V

3. Box Preparation

- Dry-fit the ICOM main unit and existing L-brackets in the empty case to confirm the 30 mm front / 35 mm back margin and left/right wall clearances before cutting anything.
- Remove the left and right internal parts-cup ribs with an oscillating multi-tool (flush-cut blade). Leave the front and back ribs in place — nothing mounts to those walls. Cut slightly proud of flush and finish by hand with a file or sanding block rather than trying to hit flush in one pass.
- Dry-fit the SO-239 flange, TRRS body, PowerPole print, and fan against the cut walls before final mounting. Check for leftover rib stubs under any flange or gasket surface — even 1 mm of proud material will tilt a connector or keep a gasket from compressing evenly.
- Where a rib is being removed right where a connector will be torqued down (PowerPole print, SO-239), plan a stainless backing washer behind the nut rather than relying on the bare wall for stiffness.
- Cut the HDPE sub-plate to fit the case floor. Confirm final floor flatness once the bin-tray ribs (if any remain molded into the floor) are accounted for.

4. Step-by-Step Build

Step 1 — Sub-plate hardware

Drill and countersink four holes in the HDPE plate matching the radio L-bracket footprint. Mount the brackets to the plate from the top with flat-head M4 bolts (existing hardware from the prior go-box project). This keeps the radio-to-plate connection independent of however the plate attaches to the case, so no underside access is ever needed.

Step 2 — Plate-to-case attachment

Apply 3M Dual Lock strips to the case floor and the matching strips to the underside of the HDPE plate, positioned under the radio's footprint for the load-bearing contact area. No standoffs, no heat-set inserts into the case itself — the plate sits flush and lifts off as a unit for service.

Step 3 — Cable-tie anchors

At each planned cable run, drill two small holes close together through the plate. Cables route atop the plate; the zip tie passes down through one hole and back up through the other to cinch over the cable on top. Anchor each run close to its connector and again partway to the radio, keeping the left-side (DC/console) and right-side (antenna/speaker) runs from crossing paths.

Step 4 — Wall cutouts

With the ribs removed (Section 3), mark and cut the six wall openings per the hole schedule: SO-239 and TRRS on the right (back and front rows), PowerPole and console RJ11/12 on the left (front and back rows), vent and fan centered on their respective walls.

Step 5 — Install bulkheads

Mount each bulkhead per its dry-fit position. Install the PowerPole print and SO-239 with backing washers where a rib was removed underneath. Fit the TPU vent and fan shrouds.

Step 6 — Wire the radio side

Run 12 AWG DC power and the antenna coax to the radio's rear panel; dress the DC leads across to the left-wall PowerPole per the original photo reference, keeping the run away from the antenna feed. Fit the ferrite choke at the DC entry point.

Step 7 — Wire the control/monitoring side

Wire the ESP32 and buck converter on their back-wall shelf: 5V rail to ESP32 and fan, PWM line from ESP32 GPIO to the fan's control pin, tach line through the divider if RPM feedback is wanted, DS18B20 data line with its pull-up, probe bonded to the heatsink.

Step 8 — Seat the sub-assembly

Press the finished radio-and-plate assembly onto the Dual Lock in the case. Connect the console, antenna, speaker, and power runs to their bulkheads. Confirm the fan and vent sit in the radio's own airflow path before closing the lid — the case fan should be pulling in the same direction the radio's fan discharges, not opposing it, with clearance between the two.

Step 9 — Final check

Power up, confirm ESP32 reads a sane temperature, fan responds to PWM commands, and TX current draw and cable dressing look correct before the box goes into the truck.

5. Still Open

[] Standoff height (reused hardware) — sets the clearance for airflow under the radio and confirms the stack still fits under the lid.

[] Filtered vent open area — media is settled (120 μm polyester monofilament mesh), but the 40×40mm frame is still provisional, matched to the fan's footprint (see the appendix template); confirm once the shroud design fixes the actual open-air passage.

[] TPU shroud dimensions for the vent and fan — not yet drawn.

6. Resources

Reference links for printed parts and other reusable designs, collected as the build progresses.

- Anderson Powerpole small panel mount — <https://www.thingiverse.com/thing:6874922>

KC1ZDJ • White Mountains, NH • Ham Radio Mobile Base Station project

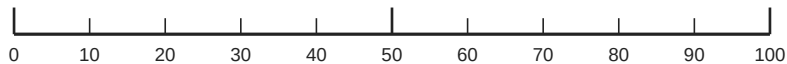
Appendix — Drilling Templates

The six templates that follow (SO-239, TRRS, the fan mounting pattern, the intake vent, the RJ11/12 console bulkhead, and the PowerPole flush panel mount) are drawn at true 1:1 scale. Print this document at 100% / Actual Size — most print dialogs default to “Fit to page,” which will rescale everything and make the holes the wrong size. Before cutting anything, measure the ruler below with a tape measure or ruler — it should read exactly 100 mm end to end. If it doesn't, your printer rescaled the page; reprint with scaling disabled.

Cut each template out along its dashed line, position it on the case wall per the wall/row assignment in Section 2 (Parts & Hole Schedule), and hold it with a small piece of double-sided tape. Use the crosshair marks to center-punch before drilling — the crosshairs mark true center, not the tape.

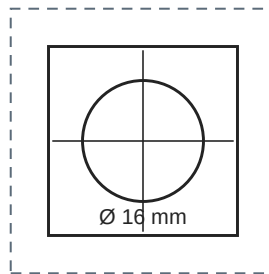
Each template also has a solid rectangular border (the trim card) with thin dotted lines extending to the page edges. Those dotted lines are alignment guides for a paper cutter — line the blade up with where a dotted line meets the edge of the sheet, cut, rotate the sheet, and repeat for all four sides to trim the rectangle out squarely. Then cut the finer dashed/round shape inside it by hand.

This appendix is padded with blank pages so every template's reverse side is blank — safe to print single-sided or duplex.



Calibration ruler — should measure exactly 100 mm

Print at 100% / Actual Size — disable “Fit to page” or “Scale to fit”

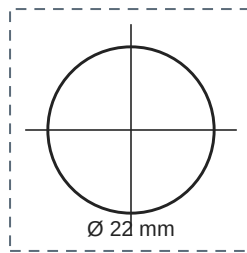


cut along dashed line

SO-239 Antenna Bulkhead

16 mm center hole · 25 × 25 mm flange
Right wall, back row

Print at 100% / Actual Size — disable “Fit to page” or “Scale to fit”

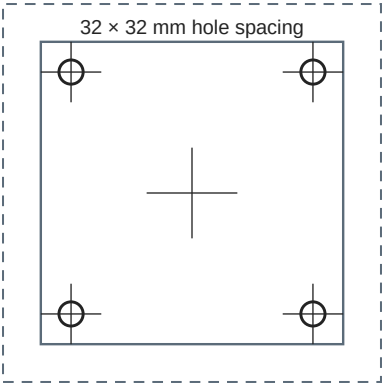


cut along dashed line

TRRS Speaker Bulkhead

22 mm diameter hole
Right wall, front row

Print at 100% / Actual Size — disable “Fit to page” or “Scale to fit”

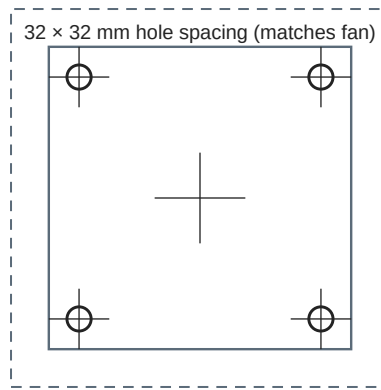


cut along dashed line

Fan Mounting — Noctua NF-A4x20 5V PWM

40 × 40 mm frame · 32 × 32 mm hole spacing · pilots shown at ~3.2 mm (typical M3 — verify)
Right wall, center. Airflow opening not shown — pending vent/shroud sizing (Section 5).

Print at 100% / Actual Size — disable “Fit to page” or “Scale to fit”



cut along dashed line

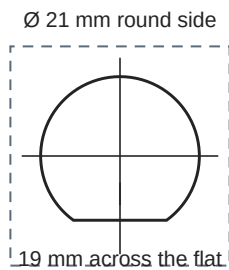
Filtered Vent — Intake

40 × 40 mm frame · 32 × 32 mm hole spacing — provisionally matched to the fan's own mounting footprint so one TPU shroud design can serve both. Confirm before finalizing — the actual open-air passage size still depends on your shroud design (Section 5).

Media: 120 µm polyester monofilament mesh, clamped against the outside face of the wall.

Left wall, center — directly opposite the fan.

Print at 100% / Actual Size — disable “Fit to page” or “Scale to fit”

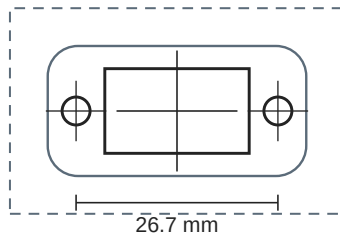


cut along dashed line

Console Bulkhead — RJ11/12

21 mm round · flat side brings it to 19 mm — drill round, then file the flat
Left wall, back row

Print at 100% / Actual Size — disable “Fit to page” or “Scale to fit”



cut along dashed line

PowerPole Bulkhead — Flush Panel Mount

19.1 × 11.2 mm through-box cutout · 2× Ø 3.7 mm holes, 26.7 mm apart

34.2 × 17.2 mm flange shown gray — reference only, not a cut line

Left wall, front row

